

Q3 Constant acceleration equations

Name: _____

Class: _____

Date: _____

Time: **425 minutes**

Marks: **352 marks**

Comments:

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Q1.

In this question use $g = 9.8 \text{ m s}^{-2}$

A ball, initially at rest, is dropped from a height of 40 m above the ground.

Calculate the speed of the ball when it reaches the ground.

Circle your answer.

-28 m s^{-1}

28 m s^{-1}

-780 m s^{-1}

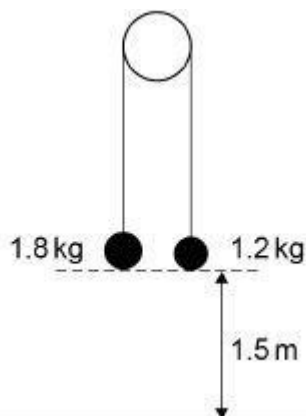
780 m s^{-1}

(Total 1 mark)

Q2.

In this question use $g = 9.8 \text{ m s}^{-2}$

Two particles, of mass 1.8 kg and 1.2 kg, are connected by a light, inextensible string over a smooth peg.



- (a) Initially the particles are held at rest 1.5 m above horizontal ground and the string between them is taut.

The particles are released from rest.

Find the time taken for the 1.8 kg particle to reach the ground.

(5)

- (b) State one assumption you have made in answering part (a).

(1)

(Total 6 marks)

Q3.

A cyclist, Laura, is travelling in a straight line on a horizontal road at a constant speed of 25 km h^{-1}

A second cyclist, Jason, is riding closely and directly behind Laura. He is also moving with a constant speed of 25 km h^{-1}

- (a) The driving force applied by Jason is likely to be less than the driving force applied by Laura.

Explain why.

(1)

- (b) Jason has a problem and stops, but Laura continues at the same constant speed.
Laura sees an accident 40 m ahead, so she stops pedalling and applies the brakes.
She experiences a total resistance force of 40 N
Laura and her cycle have a combined mass of 64 kg

- (i) Determine whether Laura stops before reaching the accident.

Fully justify your answer.

(4)

- (ii) State one assumption you have made that could affect your answer to part (b)(i).

(1)

(Total 6 marks)

Q4.

A remote-controlled toy car is moving over a horizontal surface. It moves in a straight line through a point A.

The toy is initially at the point with displacement 3 metres from A. Its velocity, v m s⁻¹, at time t seconds is defined by

$$v = 0.06(2 + t - t^2)$$

- (a) Find an expression for the displacement, r metres, of the toy from A at time t seconds.

(4)

- (b) In this question use $g = 9.8$ m s⁻²

At time $t = 2$ seconds, the toy launches a ball which travels directly upwards with initial speed 3.43 m s⁻¹

Find the time taken for the ball to reach its highest point.

(3)

(Total 7 marks)

Q5.

A garden snail moves in a straight line from rest to 1.28 cm s⁻¹, with a constant acceleration in 1.8 seconds.

Find the acceleration of the snail.

Circle your answer.

2.30 m s^{-2}

0.71 m s^{-2}

0.0071 m s^{-2}

0.023 m s^{-2}

(Total 1 mark)

Q6.

A buggy is pulling a roller-skater, in a straight line along a horizontal road, by means of a connecting rope as shown in the diagram.



The combined mass of the buggy and driver is 410 kg
 A driving force of 300 N and a total resistance force of 140 N act on the buggy.

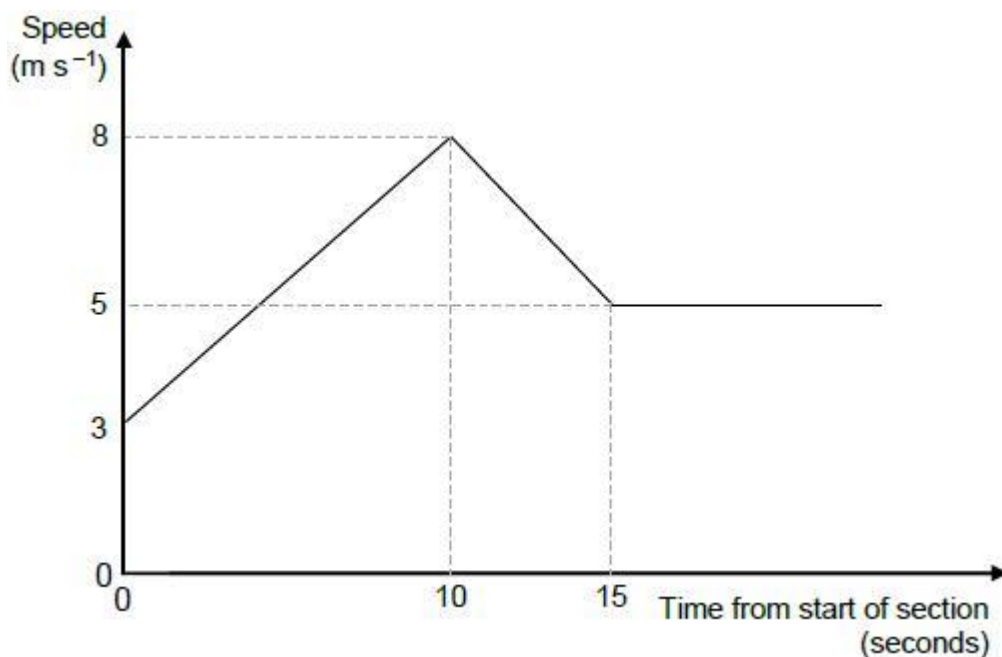
The mass of the roller-skater is 72 kg
 A total resistance force of R newtons acts on the roller-skater.

The buggy and the roller-skater have an acceleration of 0.2 m s^{-2}

- (a) (i) Find R . (3)
- (ii) Find the tension in the rope. (3)
- (b) State a necessary assumption that you have made. (1)
- (c) The roller-skater releases the rope at a point A, when she reaches a speed of 6 m s^{-1}
- She continues to move forward, experiencing the same resistance force.
- The driver notices a change in motion of the buggy, and brings it to rest at a distance of 20 m from A.
- (i) Determine whether the roller-skater will stop before reaching the stationary buggy.
 Fully justify your answer. (5)
- (ii) Explain the change in motion that the driver noticed. (2)
- (Total 14 marks)

Q7.

The graph shows how the speed of a cyclist varies during a timed section of length 120 metres along a straight track.



- (a) Find the acceleration of the cyclist during the first 10 seconds.

(1)

- (b) After the first 15 seconds, the cyclist travels at a constant speed of 5 m s^{-1} for a further T seconds to complete the 120-metre section.

Calculate the value of T .

(4)

(Total 5 marks)

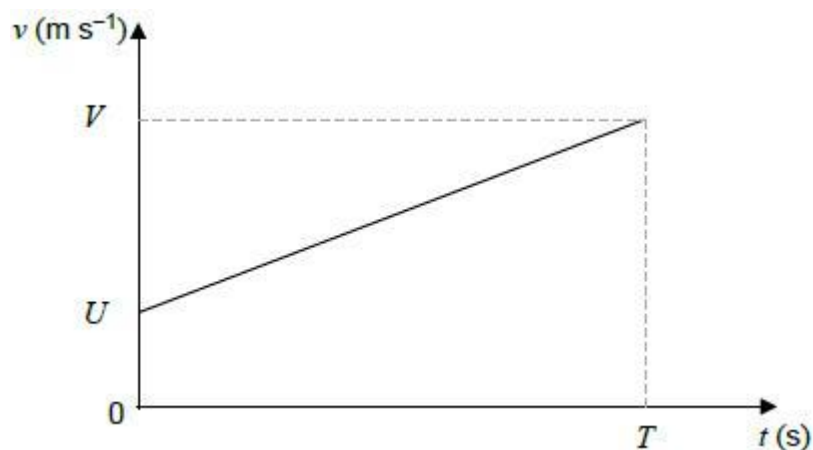
Q8.

A particle moves on a straight line with a constant acceleration, $a \text{ m s}^{-2}$.

The initial velocity of the particle is $U \text{ m s}^{-1}$

After T seconds the particle has velocity $V \text{ m s}^{-1}$

This information is shown on the velocity-time graph.



The displacement, S metres, of the particle from its initial position at time T seconds is given by the formula

$$S = \frac{1}{2}(U + V)T$$

- (a) By considering the gradient of the graph, or otherwise, write down a formula for a in terms of U , V and T .

(1)

- (b) Hence show that $V^2 = U^2 + 2aS$.

(3)

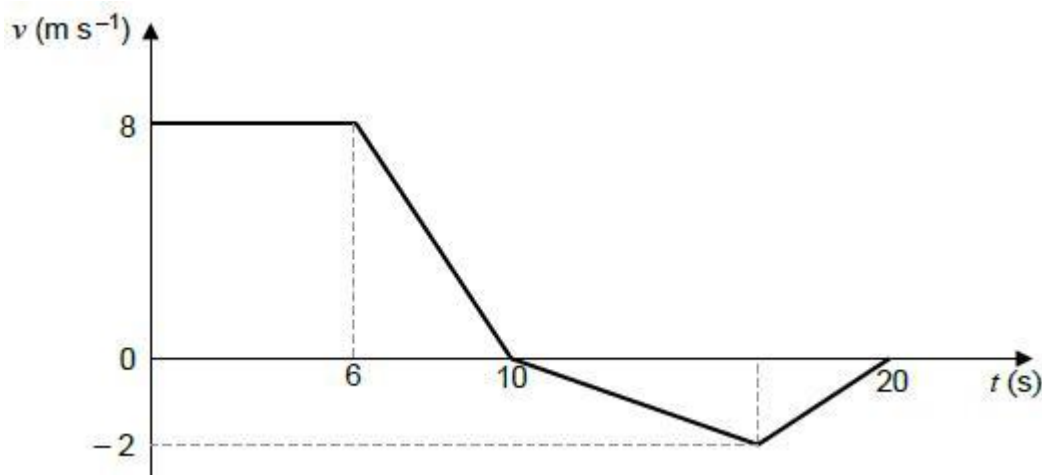
(Total 4 marks)

Q9.

The graph below models the velocity of a small train as it moves on a straight track for 20 seconds.

The front of the train is at the point A when $t = 0$

The mass of the train is 800kg.



- (a) Find the total distance travelled in the 20 seconds.

- (3)
- (b) Find the distance of the front of the train from the point A at the end of the 20 seconds. (1)
- (c) Find the maximum magnitude of the resultant force acting on the train. (2)
- (d) Explain why, in reality, the graph may not be an accurate model of the motion of the train. (1)
- (Total 7 marks)

Q10.

In this question use $g = 9.81 \text{ m s}^{-2}$, giving your final answers to an appropriate degree of accuracy.

A ball is projected from the origin. After 2.5 seconds, the ball lands at the point with position vector $(40\mathbf{i} - 10\mathbf{j})$ metres.

The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are horizontal and vertical respectively.

Assume that there are no resistance forces acting on the ball.

- (a) Find the speed of the ball when it is at a height of 3 metres above its initial position. (6)
- (b) State the speed of the ball when it is at its maximum height. (1)
- (c) Explain why the answer you found in part (b) may not be the actual speed of the ball when it is at its maximum height. (1)
- (Total 8 marks)

Q11.

A car travels on a straight horizontal race track. The car decelerates uniformly from a speed of 20 m s^{-1} to a speed of 12 m s^{-1} as it travels a distance of 640 metres.

The car then accelerates uniformly, travelling a further 1820 metres in 70 seconds.

- (a) (i) Find the time that it takes the car to travel the first 640 metres. (3)
- (ii) Find the deceleration of the car during the first 640 metres. (3)
- (b) (i) Find the acceleration of the car as it travels the further 1820 metres. (3)
- (ii) Find the speed of the car when it has completed the further 1820 metres. (3)

- (c) Find the average speed of the car as it travels the 2460 metres.

(2)

(Total 14 marks)

Q12.

A particle is initially at the point A , which has position vector $13.6\mathbf{i}$ metres, with respect to an origin O . At the point A , the particle has velocity $(6\mathbf{i} - 2.4\mathbf{j}) \text{ m s}^{-1}$, and in its subsequent motion, it has a constant acceleration of $(-0.8\mathbf{i} + 0.1\mathbf{j}) \text{ m s}^{-2}$. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) Find an expression for the velocity of the particle t seconds after it leaves A .

(2)

- (b) Find an expression for the position vector of the particle, with respect to the origin O , t seconds after it leaves A .

(3)

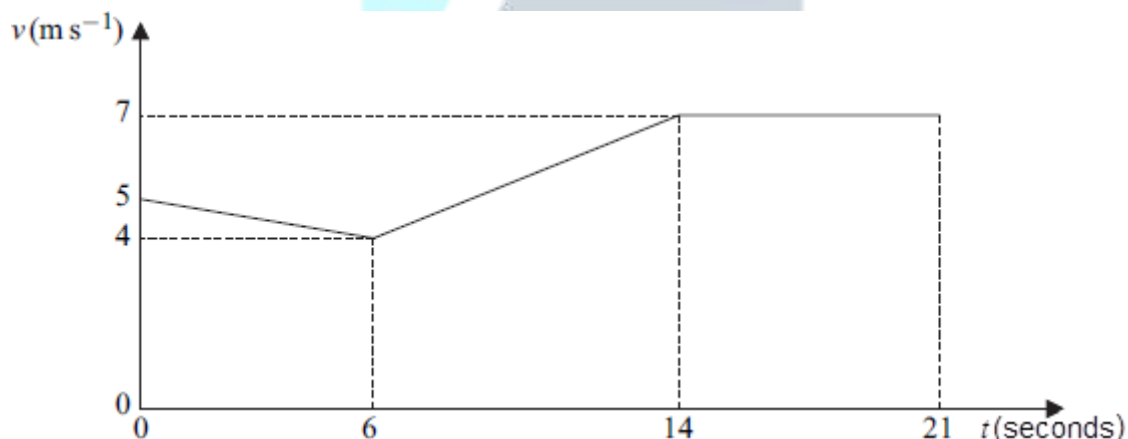
- (c) Find the distance of the particle from the origin O when it is travelling in a north-westerly direction.

(7)

(Total 12 marks)

Q13.

The graph shows how the speed of a cyclist, Hannah, varies as she travels for 21 seconds along a straight horizontal road.



- (a) Find the distance travelled by Hannah in the 21 seconds.

(4)

- (b) Find Hannah's average speed during the 21 seconds.

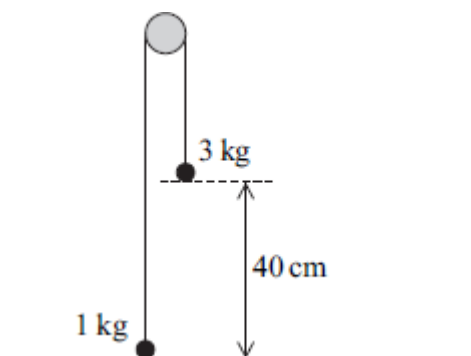
(2)

(Total 6 marks)

Q14.

Two particles are connected by a light inextensible string that passes over a smooth peg.

The particles have masses of 3 kg and 1 kg. The 1 kg particle is pulled down to ground level, where it is 40 cm below the level of the 3 kg particle, as shown in the diagram.



The particles are released from rest with the string vertical above each particle. Assume that no resistance forces act on the particles as they move.

- (a) By forming two equations of motion, one for each particle, find the magnitude of the acceleration of the particles after they have been released but before the 3 kg particle hits the ground. (5)
 - (b) Find the speed of the 1 kg particle when the 3 kg particle hits the ground. (2)
 - (c) After the 3 kg particle has hit the ground, the 1 kg particle continues to move and the string is now slack. Find the maximum height above ground level reached by the 1 kg particle. (3)
 - (d) If a constant air resistance force also acts on the particles as they move, explain how this would change your answer for the acceleration in part (a). Give a reason for your answer. (2)
- (Total 12 marks)**

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Q15.

A helicopter travels at a constant height above the sea. It passes directly over a lighthouse with position vector $(500\mathbf{i} + 200\mathbf{j})$ metres relative to the origin, with a velocity of $(-17.5\mathbf{i} - 27\mathbf{j}) \text{ m s}^{-1}$. The helicopter moves with a constant acceleration of $(0.5\mathbf{i} + 0.6\mathbf{j}) \text{ m s}^{-2}$. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) Find the position vector of the helicopter t seconds after it has passed over the lighthouse. (3)
- (b) The position vector of a rock is $(200\mathbf{i} - 400\mathbf{j})$ metres relative to the origin. Show that the helicopter passes directly over the rock, and state the time that it takes for the helicopter to move from the lighthouse to the rock. (7)
- (c) Find the average velocity of the helicopter as it moves from the lighthouse to the

rock.

(3)

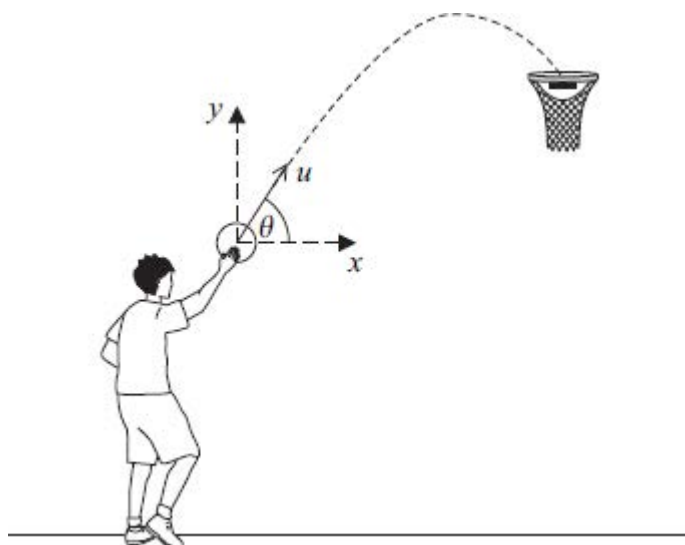
- (d) Is the magnitude of the average velocity equal to the average speed of the helicopter?
Give a reason for your answer.

(2)

(Total 15 marks)

Q16.

A player projects a basketball with speed $u \text{ m s}^{-1}$ at an angle θ above the horizontal. The basketball travels in a vertical plane through the point of projection and goes into the basket. During the motion, the horizontal and upward vertical displacements of the basketball from the point of projection are x metres and y metres respectively.



- (a) Find an expression for y in terms of x , u , g and $\tan \theta$.

(6)

- (b) The player projects the basketball with speed 8 m s^{-1} from a point 0.5 metres vertically below and 5 metres horizontally from the basket.

- (i) Show that the two possible values of θ are approximately 63.1° and 32.6° , correct to three significant figures.

(5)

- (ii) Given that the player projects the basketball at 63.1° to the horizontal, find the direction of the motion of the basketball as it enters the basket. Give your answer to the nearest degree.

(4)

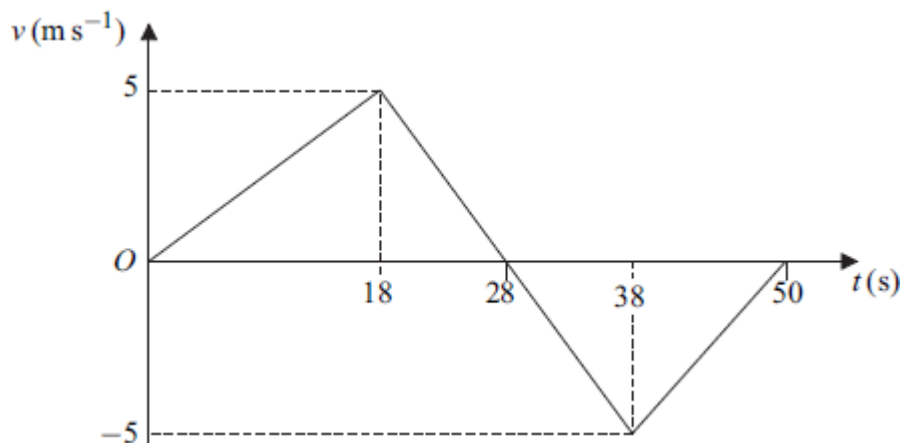
- (c) State a modelling assumption needed for answering parts (a) and (b) of this question.

(1)

(Total 16 marks)

Q17.

The diagram shows a velocity–time graph for a train as it moves on a straight horizontal track for 50 seconds.



- (a) Find the distance that the train moves in the first 28 seconds. (2)
 - (b) Calculate the total distance moved by the train during the 50 seconds. (3)
 - (c) Hence calculate the average speed of the train. (2)
 - (d) Find the displacement of the train from its initial position when it has been moving for 50 seconds. (1)
 - (e) Hence calculate the average velocity of the train. (2)
 - (f) Find the acceleration of the train in the first 18 seconds of its motion. (1)
- (Total 11 marks)**

Q18.

A cyclist freewheels, with a constant acceleration, in a straight line down a slope. As the cyclist moves 50 metres, his speed increases from 4 m s^{-1} to 10 m s^{-1} .

- (a) (i) Find the acceleration of the cyclist. (3)
- (ii) Find the time that it takes the cyclist to travel this distance. (3)
- (b) The cyclist has a mass of 70 kg. Calculate the magnitude of the resultant force acting on the cyclist. (2)

- (c) The slope is inclined at an angle α to the horizontal.
- (i) Find α if it is assumed that there is no resistance force acting on the cyclist. (3)
- (ii) Find α if it is assumed that there is a constant resistance force of magnitude 30 newtons acting on the cyclist. (3)
- (d) Make a criticism of the assumption described in part (c)(ii). (1)
- (Total 15 marks)**

Q19.

A car is travelling at a speed of 20 m s^{-1} along a straight horizontal road. The driver applies the brakes and a constant braking force acts on the car until it comes to rest. Assume that no other horizontal forces act on the car.

- (a) After the car has travelled 75 metres, its speed has reduced to 10 m s^{-1} . Find the acceleration of the car. (3)
- (b) Find the time taken for the speed of the car to reduce from 20 m s^{-1} to zero. (2)
- (c) Given that the mass of the car is 1400 kg, find the magnitude of the constant braking force. (2)
- (Total 7 marks)**

Q20.

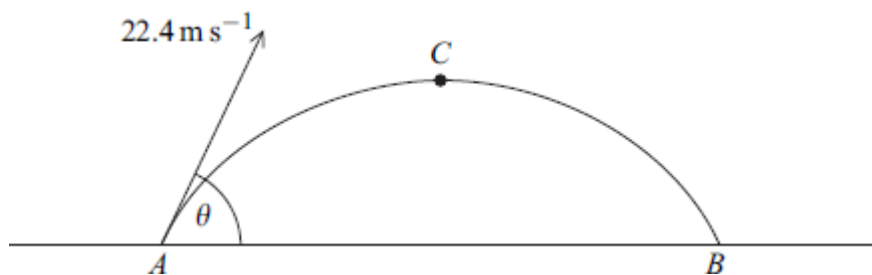
A particle moves with a constant acceleration of $(0.1\mathbf{i} - 0.2\mathbf{j}) \text{ m s}^{-2}$. It is initially at the origin where it has velocity $(-\mathbf{i} + 3\mathbf{j}) \text{ m s}^{-1}$. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) Find an expression for the position vector of the particle t seconds after it has left the origin. (2)
- (b) Find the time that it takes for the particle to reach the point where it is due east of the origin. (3)
- (c) Find the speed of the particle when it is travelling south-east. (6)
- (Total 11 marks)**

Q21.

A particle is launched from the point A on a horizontal surface, with a velocity of 22.4 m s^{-1}

at an angle θ above the horizontal, as shown in the diagram.



After 2 seconds, the particle reaches the point C , where it is at its maximum height above the surface.

- (a) Show that $\sin \theta = 0.875$. (3)
 - (b) Find the height of the point C above the horizontal surface. (3)
 - (c) The particle returns to the surface at the point B . Find the distance between A and B . (3)
 - (d) Find the length of time during which the height of the particle above the surface is greater than 5 metres. (5)
- (Total 14 marks)

Q22.

A car is travelling at a speed of 20 m s^{-1} along a straight horizontal road. The driver applies the brakes and a constant braking force acts on the car until it comes to rest.

- (a) Assume that no other horizontal forces act on the car.
 - (i) After the car has travelled 75 metres, its speed has reduced to 10 m s^{-1} . Find the acceleration of the car. (3)
 - (ii) Find the time taken for the speed of the car to reduce from 20 m s^{-1} to zero. (2)
 - (iii) Given that the mass of the car is 1400 kg, find the magnitude of the constant braking force. (2)
 - (b) Given that a constant air resistance force of magnitude 200 N acts on the car during the motion, find the magnitude of the constant braking force. (1)
- (Total 8 marks)

Q23.

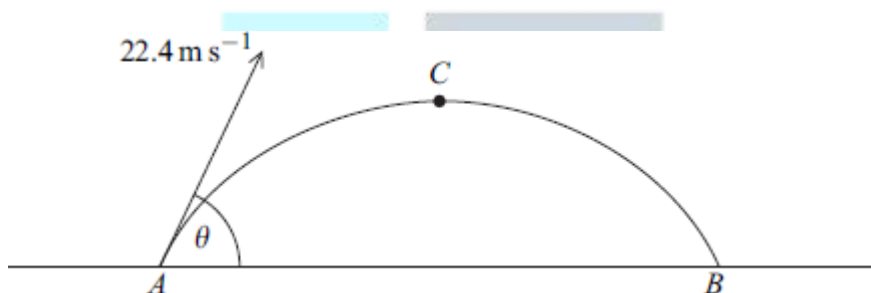
A particle moves with a constant acceleration of $(0.1\mathbf{i} - 0.2\mathbf{j}) \text{ m s}^{-2}$. It is initially at the origin where it has velocity $(-\mathbf{i} + 3\mathbf{j}) \text{ m s}^{-1}$. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) Find an expression for the position vector of the particle t seconds after it has left the origin. (2)
- (b) Find the time that it takes for the particle to reach the point where it is due east of the origin. (3)
- (c) Find the speed of the particle when it is travelling south-east. (6)

(Total 11 marks)

Q24.

A particle is launched from the point A on a horizontal surface, with a velocity of 22.4 m s^{-1} at an angle θ above the horizontal, as shown in the diagram.



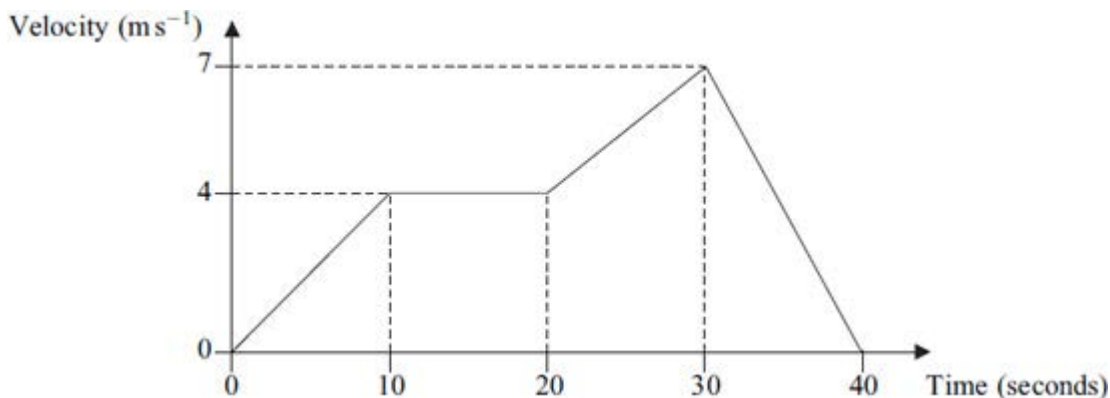
After 2 seconds, the particle reaches the point C , where it is at its maximum height above the surface.

- (a) Show that $\sin \theta = 0.875$. (3)
- (b) Find the height of the point C above the horizontal surface. (3)
- (c) The particle returns to the surface at the point B . Find the distance between A and B . (3)
- (d) Find the length of time during which the height of the particle above the surface is greater than 5 metres. (5)
- (e) Find the minimum speed of the particle. (2)

(Total 16 marks)

Q25.

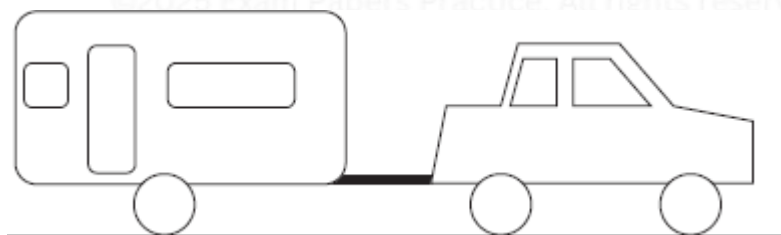
The graph shows how the velocity of a train varies as it moves along a straight railway line.



- (a) Find the total distance travelled by the train. (4)
 - (b) Find the average speed of the train. (2)
 - (c) Find the acceleration of the train during the first 10 seconds of its motion. (2)
 - (d) The mass of the train is 200 tonnes. Find the magnitude of the resultant force acting on the train during the first 10 seconds of its motion. (2)
- (Total 10 marks)**

Q26.

A car, of mass 1200 kg, tows a caravan, of mass 1000 kg, along a straight horizontal road. The caravan is attached to the car by a horizontal tow bar, as shown in the diagram.



Assume that a constant resistance force of magnitude 200 newtons acts on the car and a constant resistance force of magnitude 300 newtons acts on the caravan. A constant driving force of magnitude P newtons acts on the car in the direction of motion. The car and caravan accelerate at 0.8 m s^{-2} .

- (a) (i) Find P . (3)
- (ii) Find the magnitude of the force in the tow bar that connects the car to the caravan.

(3)

- (b) (i) Find the time that it takes for the speed of the car and caravan to increase from 7 m s^{-1} to 15 m s^{-1} .

(3)

- (ii) Find the distance that they travel in this time.

(3)

- (c) Explain why the assumption that the resistance forces are constant is unrealistic.

(1)

(Total 13 marks)

Q27.

A particle moves with constant acceleration $(-0.4\mathbf{i} + 0.2\mathbf{j}) \text{ m s}^{-2}$. Initially, it has velocity $(4\mathbf{i} + 0.5\mathbf{j}) \text{ m s}^{-1}$. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) Find an expression for the velocity of the particle at time t seconds.

(2)

- (b) (i) Find the velocity of the particle when $t = 22.5$.

(2)

- (ii) State the direction in which the particle is travelling at this time.

(1)

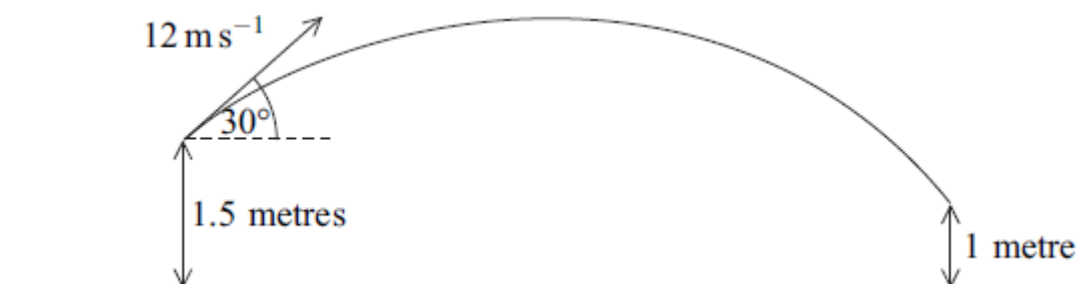
- (c) Find the time when the speed of the particle is 5 m s^{-1} .

(6)

(Total 11 marks)

Q28.

An arrow is fired from a point at a height of 1.5 metres above horizontal ground. It has an initial velocity of 12 m s^{-1} at an angle of 30° above the horizontal. The arrow hits a target at a height of 1 metre above horizontal ground. The path of the arrow is shown in the diagram.



Model the arrow as a particle.

- (a) Show that the time taken for the arrow to travel to the target is 1.30 seconds, correct to three significant figures.

(5)

- (b) Find the horizontal distance between the point where the arrow is fired and the target. (2)
- (c) Find the speed of the arrow when it hits the target. (4)
- (d) Find the angle between the velocity of the arrow and the horizontal when the arrow hits the target. (2)
- (e) State one assumption that you have made about the forces acting on the arrow. (1)
- (Total 14 marks)**

Q29.

A crane is used to lift a load, using a single vertical cable which is attached to the load. The load accelerates uniformly from rest. When it has risen 0.9 metres, its speed is 0.6 m s^{-1} .

- (a) (i) Show that the acceleration of the load is 0.2 m s^{-2} . (3)
- (ii) Find the time taken for the load to rise 0.9 metres. (2)
- (b) Given that the mass of the load is 800 kg, find the tension in the cable while the load is accelerating. (3)
- (Total 8 marks)**

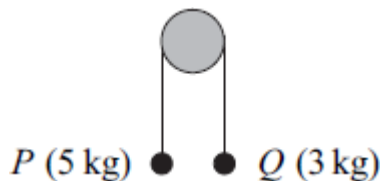
Q30.

A pair of cameras records the time that it takes a car on a motorway to travel a distance of 2000 metres. A car passes the first camera whilst travelling at 32 m s^{-1} . The car continues at this speed for 12.5 seconds and then decelerates uniformly until it passes the second camera when its speed has decreased to 18 m s^{-1} .

- (a) Calculate the distance travelled by the car in the first 12.5 seconds. (1)
- (b) Find the time for which the car is decelerating. (3)
- (c) Sketch a speed–time graph for the car on this 2000-metre stretch of motorway. (3)
- (d) Find the average speed of the car on this 2000-metre stretch of motorway. (2)
- (Total 9 marks)**

Q31.

Two particles, P and Q , are connected by a string that passes over a fixed smooth peg, as shown in the diagram. The mass of P is 5 kg and the mass of Q is 3 kg.



The particles are released from rest in the position shown.

- (a) By forming an equation of motion for each particle, show that the magnitude of the acceleration of each particle is 2.45 m s^{-2} . (5)
- (b) Find the tension in the string. (2)
- (c) State **two** modelling assumptions that you have made about the string. (2)
- (d) Particle P hits the floor when it has moved 0.196 metres and Q has not reached the peg.
 - (i) Find the time that it takes P to reach the floor. (3)
 - (ii) Find the speed of P when it hits the floor. (2)

(Total 14 marks)

Q32.

A helicopter is initially hovering above a lighthouse. It then sets off so that its acceleration is $(0.5\mathbf{i} + 0.375\mathbf{j}) \text{ m s}^{-2}$. The helicopter does not change its height above sea level as it moves. The unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) Find the speed of the helicopter 20 seconds after it leaves its position above the lighthouse. (4)
- (b) Find the bearing on which the helicopter is travelling, giving your answer to the nearest degree. (3)
- (c) The helicopter stops accelerating when it is 500 metres from its initial position.

Find the time that it takes for the helicopter to travel from its initial position to the point where it stops accelerating.

(5)
(Total 12 marks)

Q33.

(In this question, use $g = 10 \text{ m s}^{-2}$.)

A golf ball is hit from a point O on a horizontal golf course with a velocity of 40 m s^{-1} at an angle of elevation θ . The golf ball travels in a vertical plane through O . During its flight, the horizontal and upward vertical distances of the golf ball from O are x and y metres respectively.

- (a) Show that the equation of the trajectory of the golf ball during its flight is given by

$$x^2 \tan^2 \theta - 320x \tan \theta + (x^2 + 320y) = 0$$

(6)

- (b) (i) The golf ball hits the top of a tree, which has a vertical height of 8 m and is at a horizontal distance of 150 m from O .

Find the two possible values of θ .

(5)

- (ii) Which value of θ gives the shortest possible time for the golf ball to travel from O to the top of the tree? Give a reason for your choice of θ .

(2)

(Total 13 marks)

Q34.

A ball is thrown vertically upwards at a speed of 4.9 m s^{-1} from a height of 1 metre above ground level. The ball is caught when it returns to a height of 1 metre above ground level and is travelling downwards.

- (a) Find the time that it takes for the ball to reach its maximum height.

(3)

- (b) Find the maximum height of the ball above ground level.

(3)

- (c) State the total time for which the ball is moving.

(1)

- (d) State the speed of the ball when it is caught.

(1)

(Total 8 marks)

Q35.

A boat moves with constant acceleration, so that its velocity $\mathbf{v} \text{ m s}^{-1}$ at time t seconds is given by

$$\mathbf{v} = (4.5 - 0.9t)\mathbf{i} + (2 + 0.5t)\mathbf{j}$$

where the unit vectors $\mathbf{i}$ and $\mathbf{j}$ are directed east and north respectively.

- (a) Find the velocity of the boat when $t = 0$. (1)
- (b) Find the velocity of the boat when $t = 5$. (1)
- (c) State the direction in which the boat is travelling when $t = 5$. (1)
- (d) Find the acceleration of the boat. (3)
- (e) The mass of the boat is 250 kg. Find the magnitude of the resultant force acting on the boat. (3)
- (f) Draw a diagram to show the direction of the resultant force on the boat and calculate the angle between this force and the unit vector $\mathbf{i}$. (4)

(Total 13 marks)



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